

Figure 1. Reduced-model (V1) response families across a shared barrier continuation (structural-source-qualified: topology verified, not a manuscript-numeric reproduction)

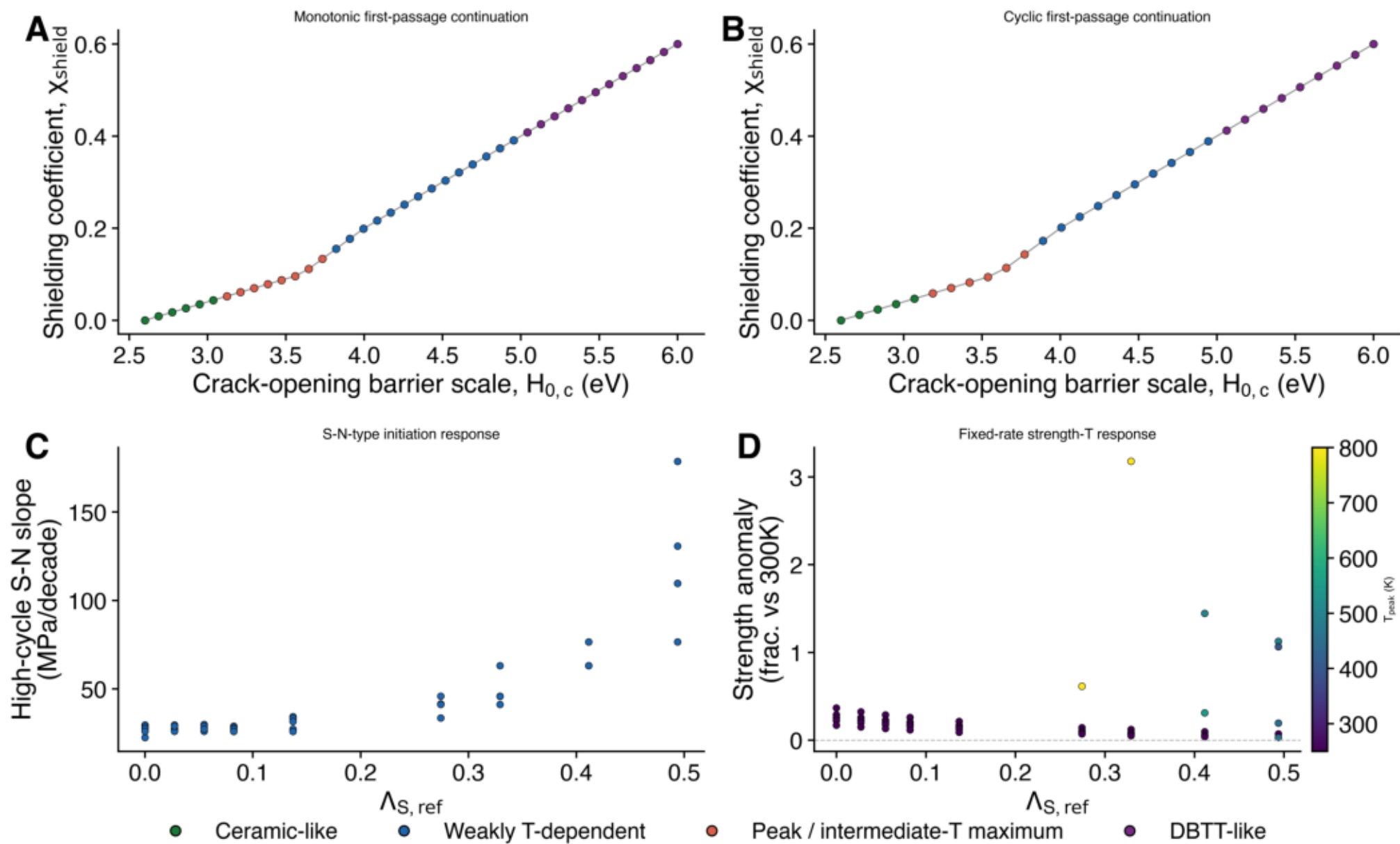


Figure 2. Matched first-passage comparison at the 1x loading rate: V1 analytical, PF/sharp-front, and FEM/CZM

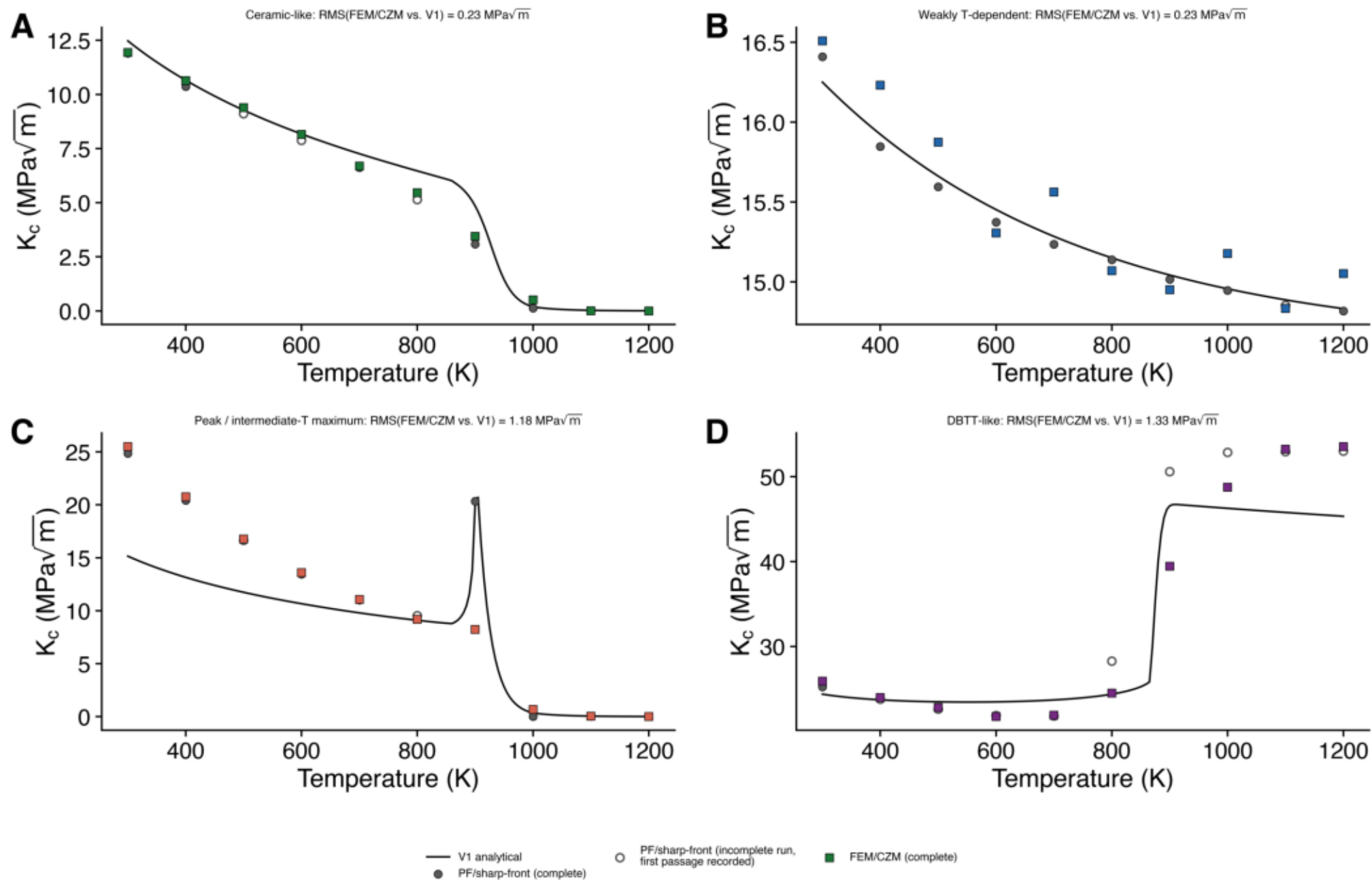


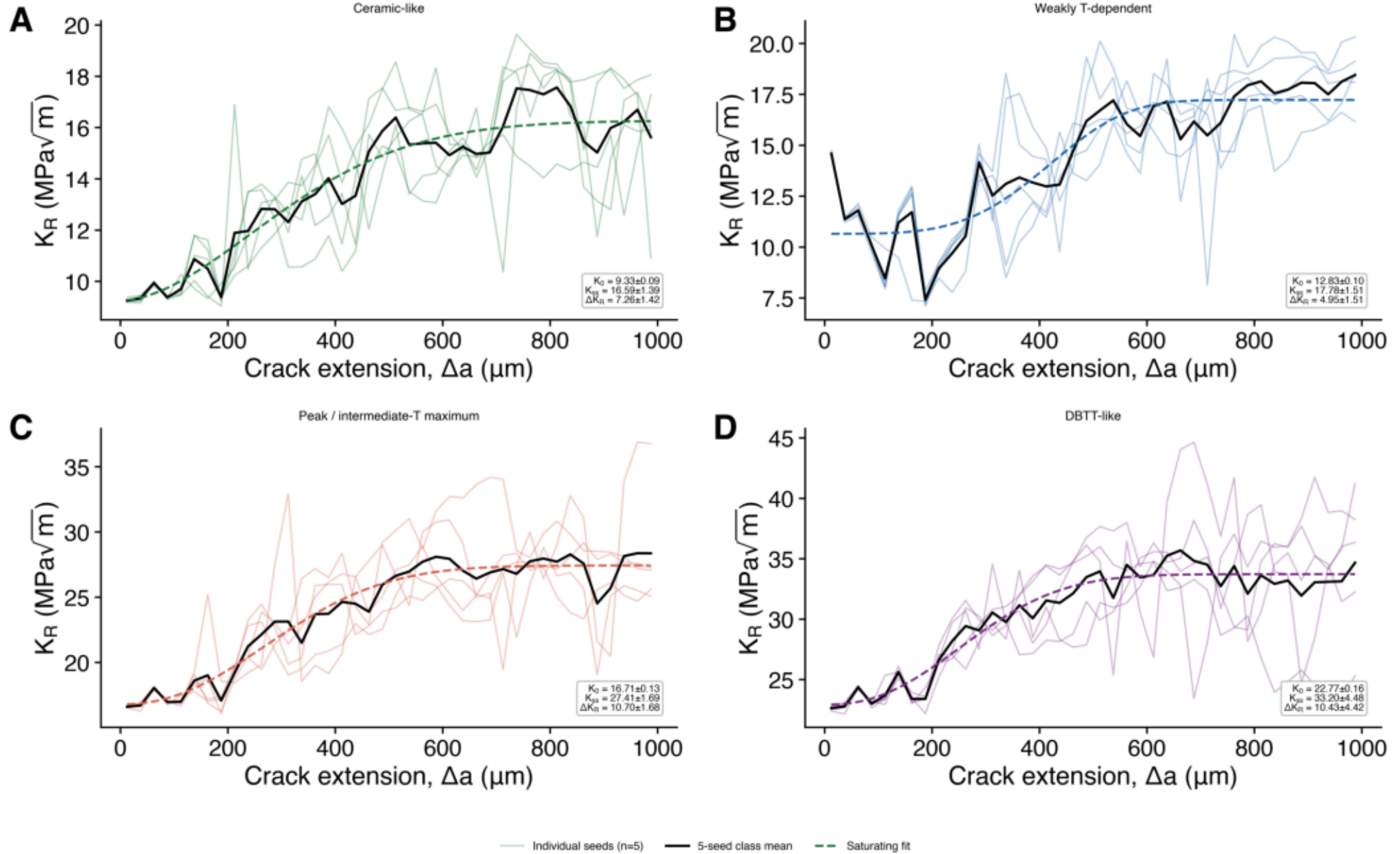
Figure 3. FEM/CZM event-sampled propagation-resistance curves at 500 K (seed-resolved; $K_R(\Delta a) = K_0 + \Delta K_R[1 - \exp\{-(\Delta a/r_R)^p\}]$)

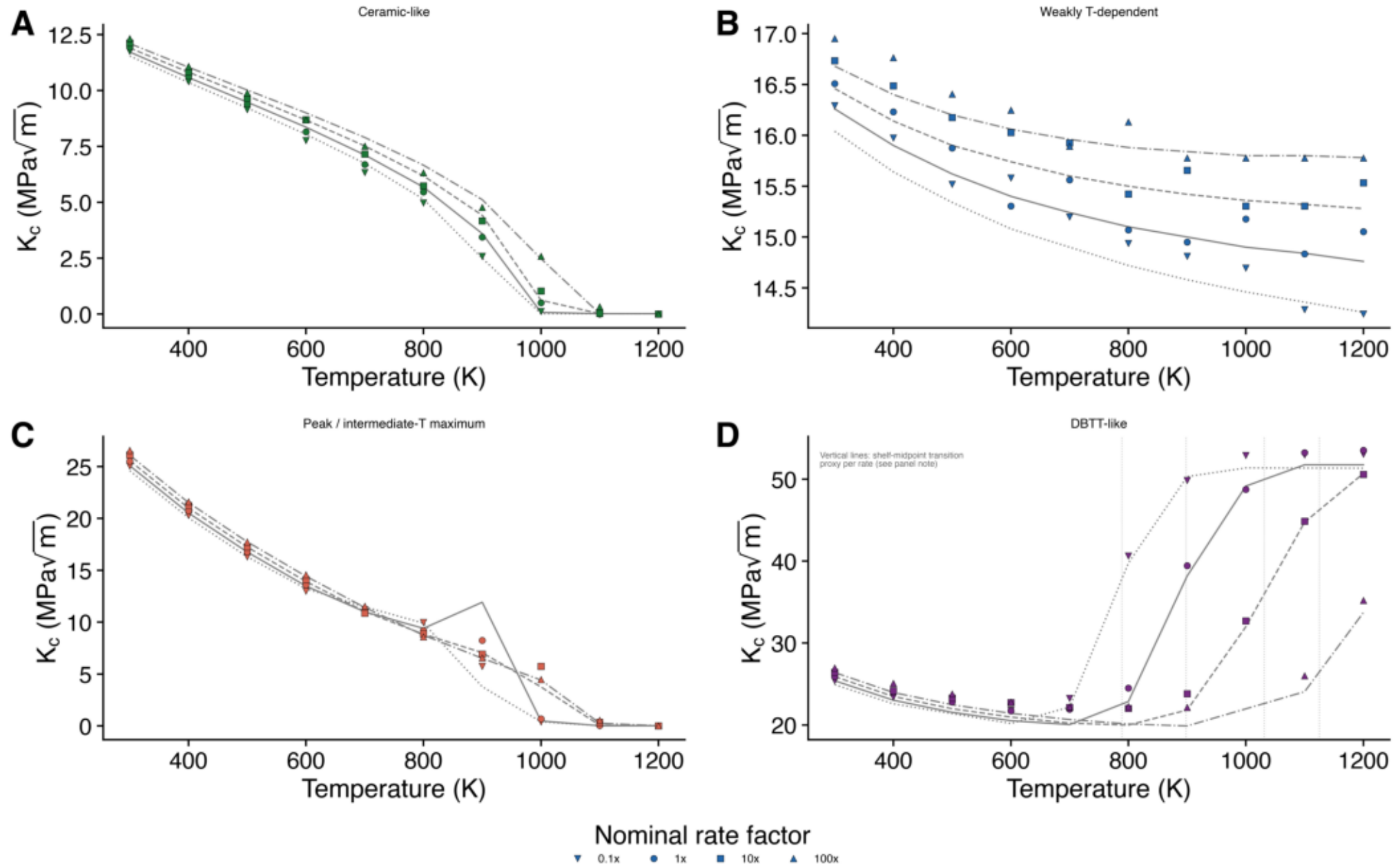
Figure 4. Loading-rate dependence of FEM/CZM first passage and the rate-matched V1 analytical prediction ($\theta=45^\circ$). FEM/CZM shows no resolved local maximum on the available temperature grid for the peak class at any rate.

Figure 5. Fatigue crack growth and crack formation generated by the shared barrier architecture

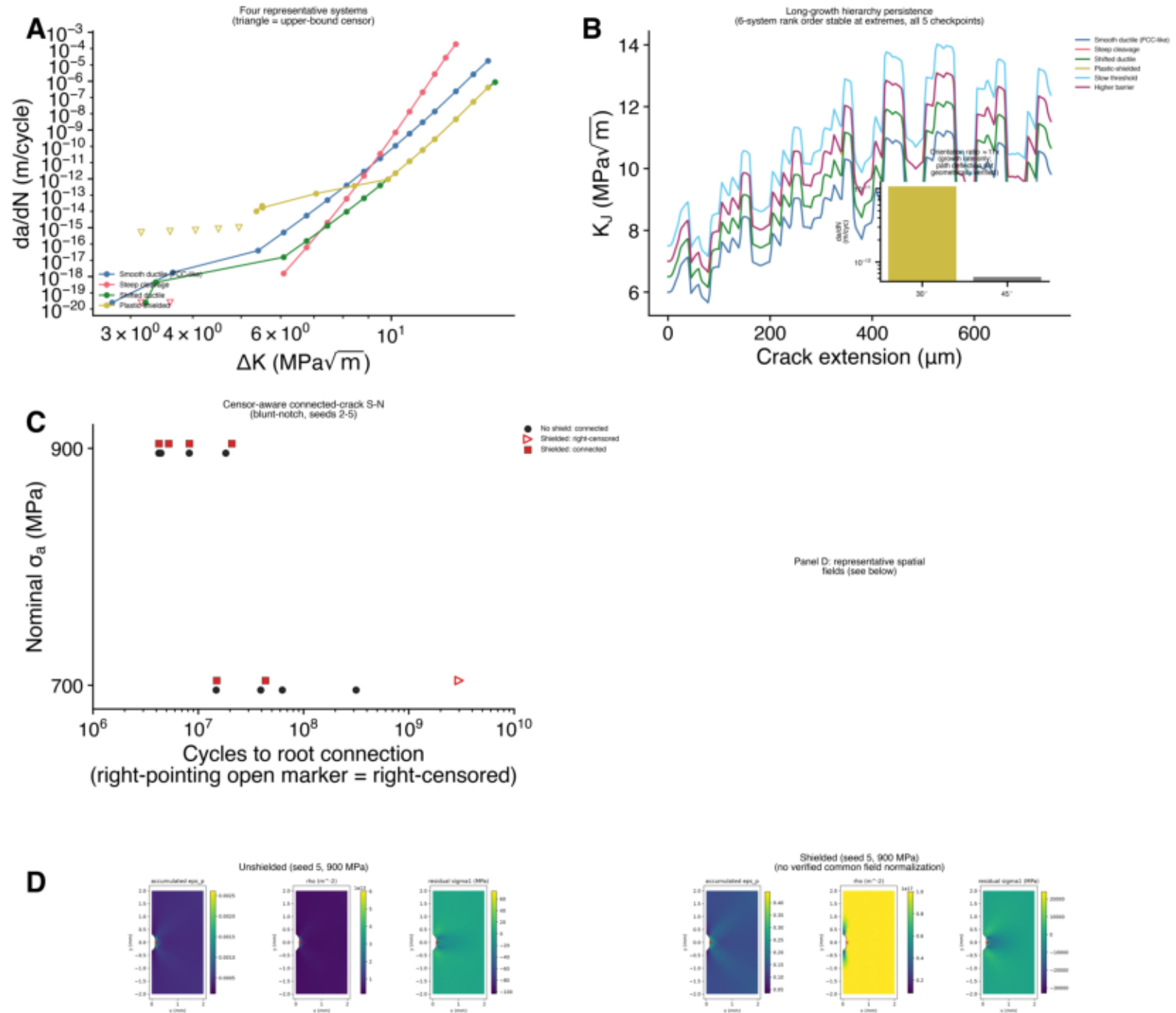


Figure 6. Cross-phenomenon relationships generated by the shared barrier landscape

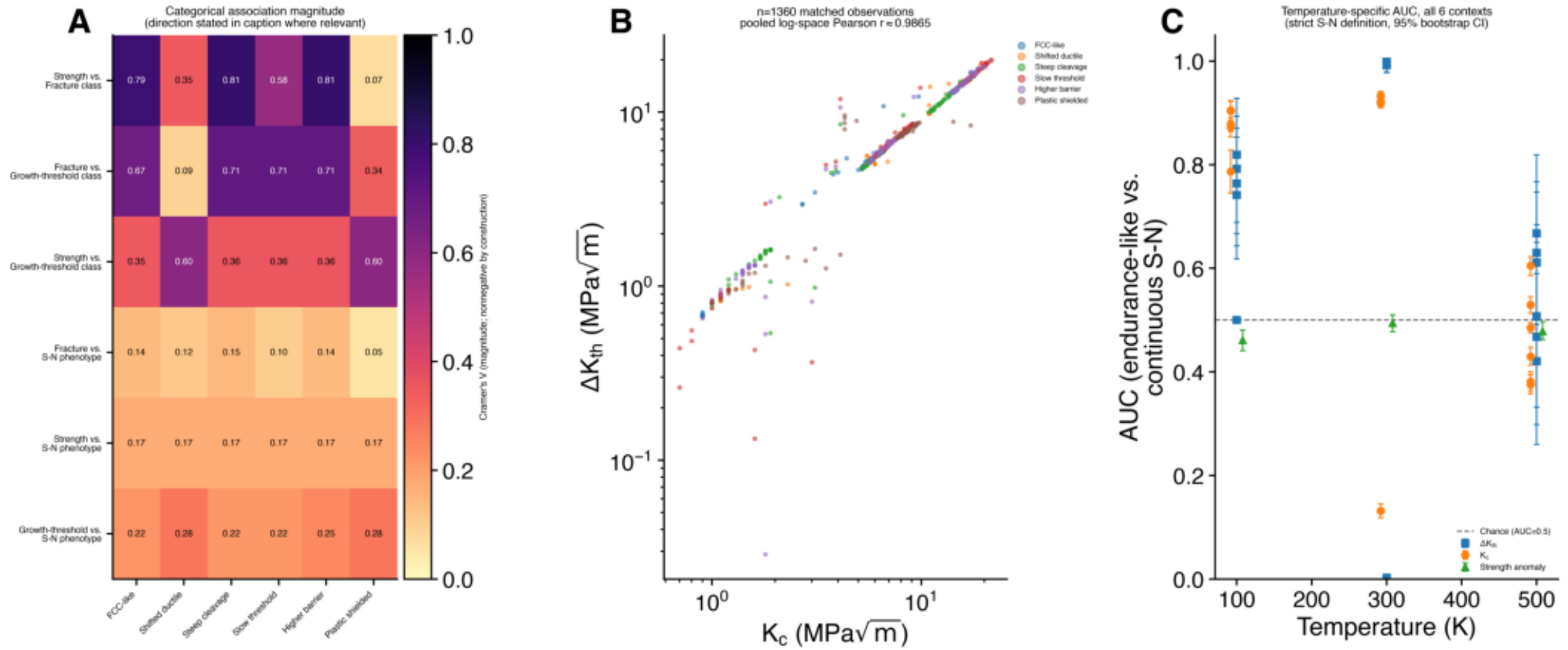


Figure 7. Cohesive-law mapping between the EXP-floor barrier form and standard cohesive representations (peak class, 900 K; fitted mapping, not a new physical simulation)

